

Design and Analysis of Algorithm

Lecture 20: Stack

Content



- **Introduction**

Question 1

The following sequence of operations is performed on stack: PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP The sequence of the value popped out is:

1. 20,10,20,10,20
2. 20,20,10,10,20
3. 10,20,20,10,20
4. 20,20,10,20,10

B

Question 2

Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order?

1. 3,4,5,1,2
2. 3,4,5,2,1
3. 1,5,2,3,4
4. 5,4,3,1,2

B

Arithmetic Notation: Transformation

1. Print operands as they arrive.
2. If the stack is empty or contains a left parenthesis on top, push the incoming operator onto the stack.
3. If the incoming symbol is a left parenthesis, push it on the stack.
4. If the incoming symbol is a right parenthesis, pop the stack and print the operators until you see a left parenthesis. Discard the pair of parentheses.
5. If the incoming symbol has higher precedence than the top of the stack, push it on the stack.
6. If the incoming symbol has equal precedence with the top of the stack, use association. If the association is left to right, pop and print the top of the stack and then push the incoming operator. If the association is right to left, push the incoming operator.
7. If the incoming symbol has lower precedence than the symbol on the top of the stack, pop the stack and print the top operator. Then test the incoming operator against the new top of stack.
8. At the end of the expression, pop and print all operators on the stack. (No parentheses should remain.)

Algorithm

Example : 1

Consider the following infix expression. Transform it into its equivalent postfix expression

$$A * (B + C * D) + E$$

	Current symbol	Operator Stack	Postfix string
1	A		A
2	*	*	A
3	(	* (	A
4	B	* (	A B
5	+	* (+	A B
6	C	* (+	A B C
7	*	* (+ *	A B C
8	D	* (+ *	A B C D
9	)	*	A B C D * +
10	+	+	A B C D * + *
11	E	+	A B C D * + * E
12			A B C D * + * E +

Question : 1

Assume that the operators $+$, $-$, $\times$ are left associative and $^$ is right associative. The order of precedence (from highest to lowest) is $^$, $\times$, $+$, $-$. What is the postfix expression corresponding to the given infix expression

$$a + b \times c - d ^ e ^ f$$

- a) $ab + c \times d - e ^ f ^$
- b) $- + a \times bc ^ ^ def$
- c) $abc \times + de ^ f ^ -$
- d) $abc \times + def ^ ^ -$

a)

Question : 2

What is the max size of operator stack during conversion of infix notation $a + b \times C - d/e$ into postfix notation

- a) 1
- b) 2
- c) 3
- d) 4

b)

Evaluation of a Postfix Expression

To evaluate a postfix expression, we can use a stack. Idea is to iterate the expression from left to right and keep on storing the operands into a stack. Once an operator is received, pop the two topmost elements and evaluate them and push the result in the stack again.

1. Add a right parenthesis ")" at the end of P.
2. [This acts as a sentinel.]
3. Scan P from left to right and repeat Steps 3 and 4 for each element of P until the sentinel ")" is encountered.
4. If an operand is encountered, put it on STACK.
5. If an operator $\otimes$ is encountered, then:
 - (a) Remove the two top elements of STACK, where A is the top element and B is the next-to-top element.
 - (b) Evaluate $B \otimes A$.
 - (c) Place the result of (b) back on STACK.
6. Set VALUE equal to the top element on STACK.
7. Exit.

Example: 1

Consider the following arithmetic expression P written in postfix notation:

$P: 5 \ 6 \ 2 \ + \ * \ 12 \ 4 \ / \ -$

	Current symbol	Stack
1	5	5
2	6	5 6
3	2	5 6 2
4	+	5 8
5	*	40
6	12	40 12
7	4	40 12 4
8	/	40 3
9	-	37
10	)	

Question: 1

Convert the following *pre-fix* expression to *in-fix*. Which one is the correct expression

$$P: - * + A B C * - D E + F G$$

- a) $(A - B) * C + D * E - (F + G)$
- b) $(A + B) * C - (D - E) * (F + G)$
- c) $(A + B - C) * (D - E) * (F + G)$
- d) $(A + B) * C - D * E - (F + G)$

$$- * + ABC * - DE \widehat{+ FG}$$

$$- * + ABC * - DE \widehat{(F + G)}$$

$$- * + ABC * \widehat{(D - E)} \widehat{(F + G)}$$

$$- * + ABC \widehat{(D - E) * (F + G)}$$

$$- * \widehat{(A + B)} C \widehat{(D - E) * (F + G)}$$

$$- \widehat{(A + B) * C} \widehat{(D - E) * (F + G)}$$

$$\widehat{(A + B) * C - (D - E) * (F + G)}$$

Question : 2

What is the max size of operator stack during evaluation of postfix expression $6\ 2\ 3\ +\ -\ 3\ 8\ 2\ \backslash\ +\ \times$

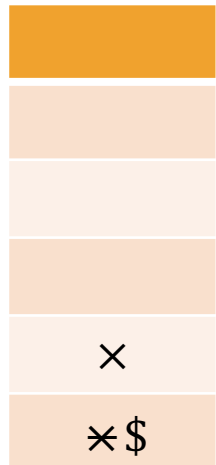
- a) 1
- b) 2
- c) 3
- d) 4

d)

Question : 3

Consider that $-$, $\times$ and $\$$ are used to indicate the operations subtraction, multiplication and exponent. Also assume that $-$ has higher precedence than $\times$, which has higher precedence than $\$$. Find the postfix notation and solve the postfix expression for the following input

- $3 - 2 \times 4 \$ 1 \times 2 \$ 3$ considering all operators are $L - R$ associative



3 2 - 4 × 1 2 × \$ 3 \$

4096